

VATTIKUTI MOHIT

Trainee Software Engineer

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🔗 HackerRank

Summary

Final-year CSE-AI student and aspiring Trainee Software Engineer who crafts working systems, not just prototypes – a from-scratch transformer with distributed (FSDP) pretraining and a batched-inference serving pipeline, alongside a userspace TCP/IP stack engineered from raw bytes in C++. Delivered measurable wins across projects – 2.36x faster inference and a 35% reduction in defect turnaround time.

Education

Pragati Engineering College, Andhra Pradesh *B.Tech in CSE – Artificial Intelligence* *Sept 2023 – Present*
GPA: 7.2/10

Projects

Userspace TCP/IP Stack (tinystack)

[demo](#) [code](#)

- Engineered a userspace TCP/IP stack in C++ with zero OS calls, handling 50 concurrent connections via raw byte manipulation and RFC 793 compliance.
- Developed a FastAPI inference server leveraging KV-caching to deliver a 2.36x speedup in LLM generation.
- Architected and deployed an ML inference server, boosting serving throughput by 2.25x through batched generation and validating outputs with exact-match correctness checks.
- Tech Stack:** C++20, CMake, ICMP, TCP, RFC 793, TUN/TAP, non-blocking I/O, RFC 1071 checksum

LLM-From-Scratch: Pretraining, Distributed Training & Serving

[demo](#) [code](#)

- Implemented a decoder-only transformer from scratch (RoPE, RMSNorm, SwiGLU, GQA) with a custom BPE tokenizer and verified multi-process FSDP pretraining via torchrun.
- Architected a FastAPI batched-inference server with KV-caching, achieving a 2.36x inference speedup and 2.25x serving throughput over unbatched generation, confirmed via exact-match correctness tests against the unbatched baseline.
- Diagnosed and fixed a DPO reward-hacking failure (100% preference accuracy masking a degenerate, collapsed model) via reward-trajectory analysis and multi-seed ablations.
- Tech Stack:** PyTorch, FastAPI, torchrun/FSDP, custom BPE, DPO

NeonAura AI – XR Hand-Tracking Platform

[demo](#) [code](#)

- Launched a zero-backend, client-side XR app (Three.js, MediaPipe Hands, WebGL) turning a standard webcam into a gesture-controlled motion platform with 10+ distinct interactions.
- Developed an Air Drawing Studio with live letter recognition and a shader-based CGI Liquid Hand with physics-driven finger deformation; trimmed frame latency by 30%+ across all gestures.
- Released six visual themes (Rainbow, Cyberpunk, Lava, Ocean, Galaxy, Plasma) and an AR mode with real-time hand-skeleton tracking – no server dependencies.
- Tech Stack:** JavaScript (ES6), Three.js, MediaPipe Hands, WebGL, Canvas API

Technical Skills

Languages: C++, Python, JavaScript, SQL, C

AI/ML: PyTorch, TensorFlow, Scikit-learn, Librosa, MediaPipe, Grad-CAM, Computer Vision, torchrun/FSDP, Custom BPE, DPO

Web & Systems: React, Node.js, HTML/CSS, REST APIs, TCP/IP, ICMP, TUN/TAP, CMake, RFC 793, RFC 1071 checksum, non-blocking I/O, Three.js, WebGL, Canvas API

Tools: Git/GitHub, Streamlit, Hugging Face, Power BI, Microsoft SQL Server, FastAPI

Achievements

- Earned a 5-star HackerRank rating across 30+ SQL challenges, accelerating intricate query execution by 40% through indexing and rewrites; cut DSA time complexity from $O(n^2)$ to $O(n \log n)$, lifting runtime performance by 30%.
- Built two full-stack applications using JavaScript, React, and SQL, minimizing bug resolution timelines by 35% via modular architecture and rigorous code reviews.

Certifications

- Oracle Agentic AI Certified Foundations Associate
- Salesforce Certified Agentforce Specialist
- Python Essentials 1 & 2